

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of B. Pharm. Examination
University Theory Examination October 2015
PH121 Pharmaceutical Biology-II

Date: 16.10.2015 , Friday

Time: 10 a.m. to 1p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. **SECTION – I** comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to **SECTION – I** are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for **SECTION – I** is 30 minutes. Answers to **SECTION – I** must be written during the first 30 minutes of the examination.
3. Answers to **SECTION – II** and **SECTION – III** are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark.

20

1. What are the repeating units of nucleic acids?
[A]phosphate molecules
[B]nucleotides
[C]bases
[D]sugar molecules
2. The enzyme which builds a mRNA strand complimentary to the DNA transcription unit is called:
[A]DNA polymerase
[B]RNA polymerase
[C]Helicase
[D]DNA ligase
3. A heritable feature is a _____ and may have two or more variants called _____.
[A] trait/characteristics
[B]character/factors
[C]character/traits
[D]trait/factors
4. In an Adult male & female body fluid is constitute between following %
[A]40 & 30 %
[B]50 & 60 %
[C]60 & 55 %
[D]70 & 40 %
5. Plasmids are what?
[A] Extrasomal RNA molecule
[B] They are viral vector
[C] Naturally occurring extrachromosomal DNA molecules
[D] All of above
6. Embryonic stem cells are derived from which cell type?
[A] Inner cell mass of blastocyte
[B] From any host cell
[C] From inner cellmass of adult cell
[D] None of above
7. What is a communicable disease?
[A] Disease that has a long duration and is not easily cured.
[B] Disease that can be spread from one person to another.
[C] Disease that is publically and scientifically understood
[D] Disease that is related to eating and drinking
8. Disease that can be spread from one person to another.
[A] Influenza
[B] Mumps

- [C] Measles
[D] All of above
9. One of the cell is not a part of Immune system.
[A] B cells
[B] TNF alfa
[C] T cells
[D] NK cells
10. Which of the following is a Pentamer antibody?
[A] Ig A
[B] Ig M
[C] Ig G
[D] Ig D
11. Anatomical fetures of Lymph nodes consists of
[A] Cells of medulla & cortex
[B] Cells around germinal center
[C] Both A & B
[D] Cells of hypothalamus
12. Follwing is not a antimicrobial substance present in the body
[A] Dermidine
[B] Defensin
[C] Cytokine
[D] Thrombocidin
13. Which of the follwing is not a function of Lymphatic system?
[A] Drain excess fluid
[B] Transport dietary lipid
[C] Protein synthesis
[D] Carry out immune response
14. One of the following is not a part of lymphatic organ or tissue
[A] Liver
[B] Spleen
[C] Thymus
[D] Lymph nodule
15. Lysozyme is present in which tissue / fluid?
[A] Urine
[B] Lacrimal tissue
[C] Blood
[D] Brest Milk
16. Antimicrobial substances of the human body are
[A] Lymphocytes, Macrophages
[B] Complement system
[C] Antimicrobial proteins
[D] All of above
17. First line of defense as immunity are
[A] Mucus & saliva
[B] Hairs
[C] Cilia
[D] All of above
18. Which of the following vitamin is essential for eyes
[A] Vitamin C
[B] Vitamin E
[C] Vitamin A
[D] Vitamin D
19. Vitamin B1 is called as:
[A] Thiamine
[B] Niacin
[C] Rationol
[D] Pyridoxin
20. Which of the following vitamins are water soluble:
[A] Vitamin A
[B] Vitamin E & K
[C] Vitamin B & C
[D] Vitamin D

SECTION – II

- Q 2** Attempt any **FOUR** of the following
- A** What is Vitamin and give classification with biological importance. 05
 - B** Explain the term Endemic, Epidemic and Pandemic. 05
 - C** Write composition and function of Lymph and lymphatic system. 05
 - D** Give basic principles of immunity and explain innate immunity and adaptive immunity 05
 - E** Write function of Spleen 05
 - F** Write an note on Immune interactions, & Immunotherapy 05

SECTION – III

- Q 3** Attempt any **FOUR** of the following
- A** Write a short note on Mendel's Laws of genetics. 05
 - B** What is Mutation? Explain DNA repair mechanism with suitable diagram. 05
 - C** Define Gene therapy. Write a short note on gene therapy for cystic fibrosis. 05
 - D** What is Hypernatremia? Write a short note on Electrolyte imbalance. 05
 - E** Write a note on application of recombinant DNA technology. 05
 - F** Give the following difference. 05
 - 1. DNA & RNA
 - 2. Intracellular fluid & Extracellular fluid
- Q 4** Attempt any **FOUR** of the following
- A** Explain DNA replication in details. 05
 - B** Define Chromosome. Explain gene expression regulation with suitable diagram. 05
 - C** Explain chemical structure of RNA in detail. 05
 - D** What is transgenesis? Write application of transgenic plant & animal. 05
 - E** Define Bacteriophage. Write a note on cloning vectors. 05
 - F** Write in detail on extracellular fluid composition. 05
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